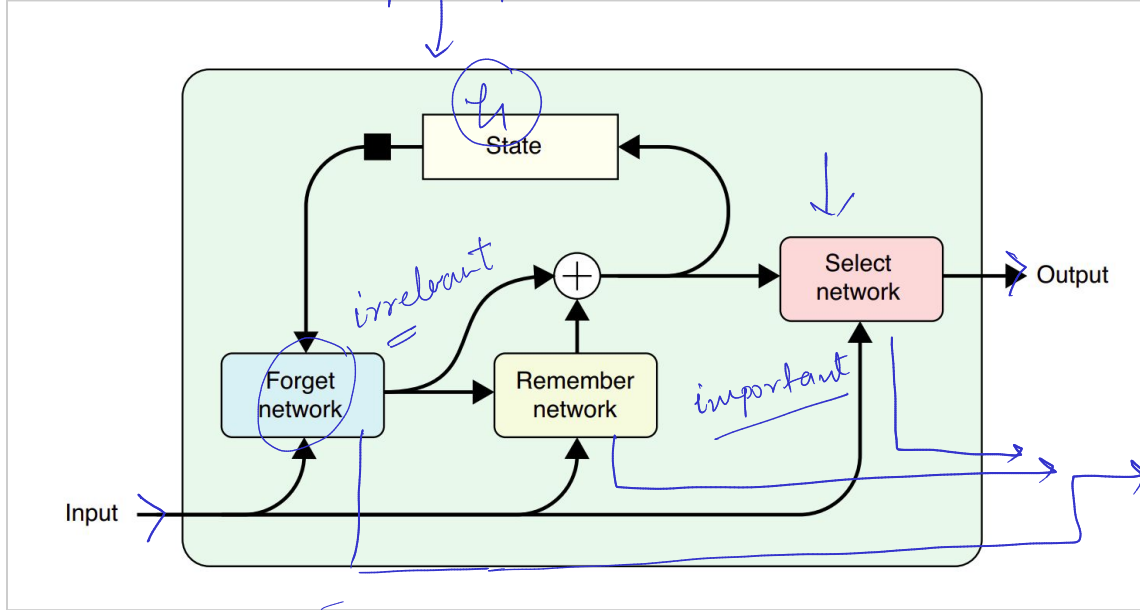
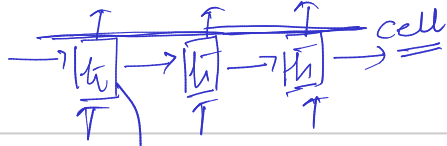


Long Short-term memory networks

($LSTM_s$) $\rightarrow$ $\boxed{\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}} \rightarrow$



Hochreiter & Schmidhuber (1997) solved the problem of **long term dependencies** by designing a memory cell using logistic and linear units with multiplicative interactions. ✓

Key idea of Long Short-term Memory

- ★ **Cell-states** – transport the information through the units → signals, gradients
 - ★ **Gates** – selectively allow information passage
- “add or remove information to cell-state using gates”**

$$(W_{hh})^n \Rightarrow \nabla \rightarrow 0$$

Cell-states have simple transformations and **eliminate the vanishing and exploding** gradient problems. Cell-state channel provides a superhighway for **uninterrupted gradient flow** and a superhighway for flow of selective (relevant) information.

